# HECTOR GENARO PACHECO

**Software Engineer | Full-Stack • Backend & APIs • Cloud Infrastructure • Mobile Systems**

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## PROFESSIONAL SUMMARY

Software Engineer and Founder at Pacific Horizon Labs LLC with hands-on software development and technical training dating to 2017, progressing from foundational web development and intensive full-stack training through university computer science coursework, collaborative engineering, containerized cloud infrastructure on AWS ECS Fargate, native mobile applications (SwiftUI & Kotlin), and commercial product delivery. Proven track record of end-to-end product ownership spanning type-safe REST APIs (Node.js/Express), relational data modeling (PostgreSQL/Prisma), human-in-the-loop AI workflows, and comprehensive automated test suites across collaborative team codebases and commercial products.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Swift, SQL, Kotlin, C++

**Frontend & Mobile:** React, Next.js 16, SwiftUI (iPadOS), Android (Kotlin), Jetpack Compose, HTML5, CSS3

**Backend & APIs:** Node.js, Express, REST APIs, GraphQL, FastAPI, Zod, RBAC

**Quality & Testing:** Vitest, Unit & Integration Testing, Automated Testing, Audit Logging

**Cloud & DevOps:** AWS (ECS Fargate, ECR, RDS, Secrets Mgr), Docker, Linux, Git, GitLab, Jira, Vercel

**Data & Databases:** PostgreSQL, Prisma ORM, Supabase, Relational Schema Design

**AI & Automation:** Human-in-the-Loop AI, Twilio Voice API, Claude API, Palantir Foundry / AIP

**Systems & Hardware:** Raspberry Pi 5, AVFoundation, Touchscreen, systemd, GPIO, 4–20 mA, PLC

## PROFESSIONAL & TECHNICAL EXPERIENCE

**Pacific Horizon Labs LLC** — *Founder & Software Engineer* COMPANY SITE ↗ PHOTO BOOTH ↗ RISING OPS ↗ **2026 – Present**

- Founded commercial software entity developing production event and operations systems; engineered Pacific Horizon Photo Booth, advancing an active-cooled Raspberry Pi 5 / Linux hardware prototype into a native SwiftUI / iPadOS product with AVFoundation capture and Supabase cloud delivery.
- Architected Pacific Rising Ops, a business operations system integrating Twilio Voice and Gmail APIs with a deterministic action state machine and Zod validation, strictly requiring human-in-the-loop approval before protected external actions.
- Own end-to-end engineering across product architecture, type-safe REST APIs, native mobile software, automated workflow pipelines, and customer-facing business systems.

**California State University, Fullerton** — *Instructional Student Assistant / Grader (Computer Science)* **Jan 2025 – Dec 2025**

- Mentored and evaluated 150+ students in introductory computer science (CPSC 120 / C++), diagnosing compiler errors, explaining data structures, and guiding students through Git/GitHub workflows.

**Software Engineering Contributor** — *Nonprofit serving underserved communities* **2025 – 2026**

- Collaborated on an existing nonprofit software platform using Jira-driven ticket workflows, Git/GitLab version control, REST API investigation, and team code debugging in a shared codebase.

**Instrumentation & Controls Technician** — *Process Control & Industrial Systems* **Field Engineering Experience**

- Calibrated, commissioned, and troubleshot 4–20 mA process instrumentation (pressure, temperature, flow, level), PLC I/O wiring, relays, and industrial power loops across 480VAC, 120VAC, and 24VDC systems with systematic fault isolation and auditable logs.

## FEATURED SOFTWARE ENGINEERING PROJECTS

**CalTrack** — *Industrial Calibration SaaS & Mobile Platform* **React • TypeScript • Node.js • Express • PostgreSQL • Docker • AWS ECS Fargate • Kotlin / Android** (Jun 2026 – Jul 2026) LIVE APP ↗ GITHUB ↗

- Architected and deployed a full-stack industrial calibration SaaS platform managing instrument tags, calibration logs, RBAC, and auditable history snapshots using Prisma ORM and PostgreSQL.
- Containerized Node/Express REST API with Docker on AWS ECS Fargate with RDS PostgreSQL in isolated VPC subnets, automated ECR image deployment, and Secrets Manager integration.
- Engineered a native Android field companion application in Kotlin with Jetpack Compose, ViewModels, CameraX, and ML Kit barcode scanning.

**EstateOS** — *Agentic Real Estate Investment Intelligence Platform* (Aug 2026 – Sep 2026) **Next.js 16 • TypeScript • Claude API • Multi-Agent • PostgreSQL • Prisma • Vitest** CASE STUDY ↗

- Built an agentic real estate intelligence platform in Next.js 16 and TypeScript, combining Claude-powered research orchestration with deterministic underwriting calculations.
- Implemented capital-stack analysis, LTV/CLTV thresholds, cash flow, debt/equity evaluation, multi-strategy screening, and provenance-aware evidence verification backed by PostgreSQL and Prisma ORM.
- Protected underwriting calculations, agent orchestration workflows, and database repository transactions with comprehensive automated Vitest test suites.

**Additional Repositories & Deployments:** **FieldTrack AI** — Edge Telemetry (Raspberry Pi/Python/React) GITHUB ↗ • **Procurement Tracking** — Full-Stack GraphQL App (React/Apollo/Prisma/PostgreSQL) LIVE APP ↗ GITHUB ↗ • **OC Water Features** — Production Web (Next.js/Supabase/Vercel) LIVE SITE ↗

## EDUCATION & PROFESSIONAL TRAINING

**The Last Mile** — Full-Stack Software Development Program (2017 – 2021) — Intensive curriculum progressing from semantic HTML/CSS and responsive design through JavaScript DOM programming, React, Node.js/Express, REST APIs, MongoDB, and Git

**Western Governors University (WGU)** — B.S. Software Engineering (In Progress, 2026–Present)

**California State University, Fullerton** — B.S. Computer Science Coursework / Project Rebound Scholar (2024–2025)

**Palantir Technologies & Mirion** — Foundry & AIP Credentials (May 2026) • SU-890 iCAM Certification